Instructions for upgrading Daewoo CPC-400/400S into the standard MSX2/MSX2+ computer

Document revision: 1.01

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April 16, 2026

This document provides detailed instructions on how to modify the Korean Daewoo CPC-400/400S computer to make it more compatible with various software using the Japanese-Russian BIOS version. The document describes several options for converting it into a more adequate MSX2 variant, as well as upgrading this computer to a full-fledged MSX2+ with an F4 port, an FMPAC Light sound card, and two megabytes of external RAM. Most of the options involve removing the onboard RAM, so these instructions are intended for users with some experience.

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This document is intended for those who are willing to perform this upgrade **at their own risk**. The author is not responsible for any issues that may arise from third parties upgrading the computer. There may be some inaccuracies in the instructions, so if you notice any, please contact the author. Contact information is provided at the end of the main document.

1. Modification Options

Several options for BIOS files and firmware have been prepared for the modification. We will list the upgrade options, from the simplest to the most complex.

- **[A]** MSX2 minimal upgrade without removing RAM
- **[B]** MSX2 medium upgrade with external RAM (SUBROM remains in slot 0.3), slot 3 is not expanded
- **[C]** MSX2 full upgrade with external boards (**recommended**)
- **[D]** MSX2+ full upgrade with external boards (**recommended**)

The first option [A] allows you to simply replace two chips, resulting in a more or less compatible computer without the Korean modules. However, some programs may not recognize the RAM in slot 0.2 and may not function properly. On the other hand, you won't have to solder anything. The other options involve removing the onboard RAM and installing additional boards in the rear slot. The project repository contains all the necessary files—ROM and PLD firmware, schematics and Gerber files for the required boards, CPLD firmware for these boards, as well as 3D models of the enclosures for all expansion boards. The first version of the RAM board and modified firmware for the slot expander are also available, which prevents this module from conflicting with the onboard Daewoo mapper.

It is important to note that when using the U33.JED firmware for the U33 programmable logic chip, the disk ROM will be located in slot 0.3, and the built-in 3.5-inch floppy drive will have priority in the system. However, if you plan to use the computer with other disk devices, such as the Carnivore2 or 2+ multi-cartridge, some boot priority issues may arise. That is, the MSX-DOS system will attempt to boot not from the cartridge's memory card, but from a floppy disk, which may cause some inconvenience—for example, resulting in DOS1 booting instead of DOS2 (Nextor). In this case, it is recommended to leave the floppy drive where it was originally—in the non-expanded slot 2. To do this, replace the U33.JED file in the U33 programmable logic chip with the **U33_DSL2.JED** file. As a result, the boot priority will be assigned to the Carnivore2 or 2+ cartridge in slot 1.

2. IMPORTANT INFORMATION!

- All work must be performed with the computer powered off
- It is extremely important to clean corrosion off the board and replace rotted wires and components
- It is recommended to install IC sockets in place of desoldered or cut-out RAM chips
- Cut-out RAM chip's pins must be removed one by one using a soldering iron
- The U38 and U44 ROM chips can be replaced with any UV-erasable EPROM chips of the 27C512 type, or with electrically erasable Winbond W27C512 EEPROM chips. The same 64 KB ROM chips are installed on external boards (slot expander, board with a standard MSX slot)
- The PAL16L8A programmable logic chips can be replaced with equivalents, such as the Lattice GAL16V8D, or similar chips from Microchip Technology (ATF16V8B).
- A standard diode, such as a 1N4148, can be used to connect the video processor to the Z80. It is installed **with the anode connected to the Z80, and it is recommended to wrap it in heat-shrink tubing** after installation. This diode connects pin 24 of the Z80 processor to pin 26 of the v9958 video processor; this is the processor's WAIT signal (video: https://youtu.be/dz-oaFYrFdg?t=1072)
- The wires on the back of the board are soldered between pins 4, 20, and 27 ("ground"), as well as between pins 21 and 58 (5V power supply) of the v9958 video processor (video: https://youtu.be/dz-oaFYrFdg?t=1015)

- If an EPROM chip is installed on the board with the MSX slot, you cannot insert a cartridge into that slot, otherwise there may be a conflict between the two different ROMs

3. Instructions for modification option [A]

This is the simplest modification option, which does not require removing the onboard RAM. Potential incompatibility with some programs may persist due to the non-standard RAM layout.

1. Unplug the computer from the power source and monitor, and disconnect the keyboard
2. Carefully remove the top cover of the case
3. Carefully remove the U38 and U44 ROM chips from their sockets
4. Carefully remove the U33 and (optionally) U46 programmable logic chips from their sockets
5. Write the U38.BIN file from the repository folder \BIOSes\Config_A\ into the 27C512 chip using a programmer
6. Install the programmed chip into the U38 socket
7. (Optional) Write the U44_???.BIN file of your choice (King's Valley game or RAM test) from the repository folder \BIOSes\Optional_U44\ into the 27C512 chip using a programmer
8. (Optional) Install the programmed chip into the U44 socket
9. Write the U33.JED or U33_DSL2.JED file from the repository folder \GALs\ into the GAL16V8D chip using a programmer
10. Install the programmed chip into the U33 panel
11. (Optional) Write the U46.BIN file from the \GALs\ repository folder into the GAL16V8D chip using a programmer
12. (Optional) Install the programmed chip into the U46 panel
13. Close the top cover of the computer case
14. Turn on the computer and check that it is working

4. Instructions for modification option [B]

This option will require, at a minimum, the removal of the onboard RAM as well as the RAM board (version 1.1). Since all ROMs remain in expanded slot 0, potential incompatibility with certain programs may persist.

1. Unplug the computer from the power source and monitor, and disconnect the keyboard
2. Carefully remove the top case cover
3. Disconnect all cables and remove the motherboard from the case
4. Carefully desolder or cut out the 4 RAM chips with wire cutters: U18, U19, U20, U21
5. Use a soldering iron to remove the cut-out pins of the chips from the board
6. (Optional) Solder the IC sockets back into the locations where the RAM chips were removed
7. Carefully remove the ROM chips U38 and U44 from their sockets
8. Carefully remove the programmable logic chips U33 and (optionally) U46 from their sockets
9. Write the U38.BIN file from the repository folder \BIOSes\Config_B\ into the 27C512 chip using a programmer
10. Install the programmed chip into the U38 socket
11. (Optional) Write the U44_???.BIN file of your choice (King's Valley game or RAM test) from the repository folder \BIOSes\Optional_U44\ into the 27C512 chip using a programmer
12. (Optional) Install the programmed chip into the U44 socket
13. Write the U33.JED or U33_DSL2.JED file from the repository folder \GALs\ into the GAL16V8D chip using a programmer
14. Install the programmed chip into the U33 panel
15. (Optional) Write the U46.BIN file from the \GALs\ repository folder into the GAL16V8D chip using a programmer
16. (Optional) Install the programmed chip into the U46 panel

17. Reinstall the motherboard into the case and connect all wires to it
18. Close the top cover of the computer case
19. Turn on the computer and check that it works

5. Instructions for modification option [C]

This modification requires removing the onboard RAM, as well as the mandatory presence of the slot expansion and RAM expansion boards. The FMPAC board is optional. With this configuration, BOTH boards must always be installed in the computer, and it is recommended to install the RAM board in the rear slot (slot 3.2). This is the most compatible configuration, which is very similar to the Yamaha MSX2 YIS-805 configuration. The only differences are the size of the RAM in slot 3.2 and the location of the disk ROM (if using the U33.JED firmware).

1. Unplug the computer from the power outlet and monitor, and disconnect the keyboard
2. Carefully remove the top case cover
3. Disconnect all cables and remove the motherboard from the case
4. Carefully desolder or cut out the 4 RAM chips with wire cutters: U18, U19, U20, U21
5. Use a soldering iron to remove the cut-out pins of the chips from the board
6. (Optional) Solder the IC sockets back into the locations where the RAM chips were removed
7. Carefully remove the ROM chips U38 and U44 from their sockets
8. Carefully remove the programmable logic chips U33 and (optionally) U46 from their sockets
9. Write the U38.BIN file from the repository folder \BIOSes\Config_C\ into the 27C512 chip using a programmer
10. Install the programmed chip into the U38 socket
11. Write the EPROM.BIN file from the repository folder \BIOSes\Config_C\ into the 27C512 chip using a programmer
12. Install the programmed chip into the expansion slot board panel
13. (Optional) Write the U44_???.BIN file of your choice (King's Valley game or RAM test) from the repository folder \BIOSes\Optional_U44\ into the 27C512 chip using a programmer
14. (Optional) Install the programmed chip into the U44 slot
15. (Optional) Use a programmer to write the FMBIOS.BIN file from the repository folder \Boards\FMPAC\ROM\ into the 27C512 chip
16. (Optional) Install the programmed chip into the FMPAC board panel
17. Write the U33.JED or U33_DSL2.JED file from the repository folder \GALs\ into the GAL16V8D chip using a programmer
18. Install the programmed chip into the U33 socket
19. (Optional) Write the U46.BIN file from the \GALs\ repository folder into the GAL16V8D chip using a programmer
20. (Optional) Install the programmed chip into the U46 panel
21. Reinstall the motherboard into the case and connect all wires to it
22. Close the top cover of the computer case
23. Connect the slot expansion board to the rear Daewoo connector
24. Insert the RAM board into the far slot expansion connector
25. (Optional) Insert the FMPAC board into the front slot expansion connector
26. Turn on the computer and check that it is working properly

6. Instructions for modification option [D]

This modification requires removing the onboard RAM, as well as the mandatory presence of the slot expansion and RAM boards. The presence of the FMPAC board is highly recommended but not required. With this configuration, BOTH boards must always be inserted into the computer, and it is recommended to insert the RAM board into the far slot (slot 3.2). This is the most advanced modification of the computer, on which all games written for MSX2+ computers will run, such as MKID and Sonyc. Such computers were manufactured only by Sanyo,

Sony, and Panasonic. However, recently MSX2+ modifications of Yamaha computers have appeared; for example, the RBSC group (https://rbsc.su) modified the YIS-503 and YIS-805 computers, and the Arabic version of the YIS-503 was also converted to MSX2+. This configuration is very similar to that of factory-made MSX2+ computers. The only differences are in the size of the RAM in slot 3.2 and the location of the disk ROM (if using the U33.JED firmware).

1. Unplug the computer from the power supply and monitor, and disconnect the keyboard
2. Carefully remove the top case cover
3. Disconnect all wires and remove the motherboard from the case
4. Carefully desolder or cut out the 4 RAM chips with wire cutters: U18, U19, U20, U21
5. Use a soldering iron to remove the cut-out pins of the chips from the board
6. (Optional) Solder the IC sockets back into the locations where the RAM chips were removed
7. Carefully remove the ROM chips U38 and U44 from their sockets
8. Carefully remove the programmable logic chips U33 and (optionally) U46 from their sockets
9. Write the U38.BIN file from the repository folder \BIOSes\Config_D\ into the 27C512 chip using a programmer
10. Install the programmed chip into the U38 socket
11. Write the EPROM.BIN file from the repository folder \BIOSes\Config_D\ into the 27C512 chip using a programmer
12. Install the programmed chip into the expansion slot board panel
13. (Optional) Write the U44_???.BIN file of your choice (King's Valley game or RAM test) from the repository folder \BIOSes\Optional_U44\ into the 27C512 chip using a programmer
14. (Optional) Install the programmed chip into the U44 socket
15. Write the FMBIOS.BIN file from the repository folder \Boards\FMPAC\ROM\ into the 27C512 chip using a programmer
16. Install the programmed chip into the FMPAC board panel
17. Write the U33.JED or U33_DSL2.JED file from the repository folder \GALs\ into the GAL16V8D chip using a programmer
18. Install the programmed chip into the U33 socket
19. (Optional) Write the U46.BIN file from the \GALs\ repository folder into the GAL16V8D chip using a programmer
20. (Optional) Install the programmed chip into the U46 panel
21. Carefully remove the v9938 video processor from socket U13
22. Flip the board over and solder jumpers between video processor pins 4, 20, and 27 (see photo 4)
23. Solder a jumper between video processor pins 21 and 58 (see photo 4)
24. Solder a 1N4148 diode with its anode connected to pin 24 of the Z80 processor (see the photo 4)
25. Solder a wire from the diode’s cathode to pin 26 of the video processor; first, slip a 2 cm length of heat shrink tubing onto the wire (see photo)
26. Slide the heat shrink tubing onto the soldered diode and heat it with a heat gun
27. Carefully install the v9958 video processor into the U13 socket, ensuring the correct orientation (the "key" on the video processor's housing must be facing the crystal oscillator
28. Reinstall the motherboard into the case and connect all wires to it
29. Close the top cover of the computer case
30. Connect the slot expansion card to the rear Daewoo connector
31. Insert the RAM board into the far slot expansion connector
32. Insert the FMPAC board into the near slot expansion connector
33. Turn on the computer and check that it is working

You can use special thermal adhesive to attach a small heat sink to the v9958 graphics processor, but this is not necessary. There is enough space inside the Daewoo case to ensure adequate cooling of all components. Furthermore, the v9958 graphics processor does not get very hot during operation.

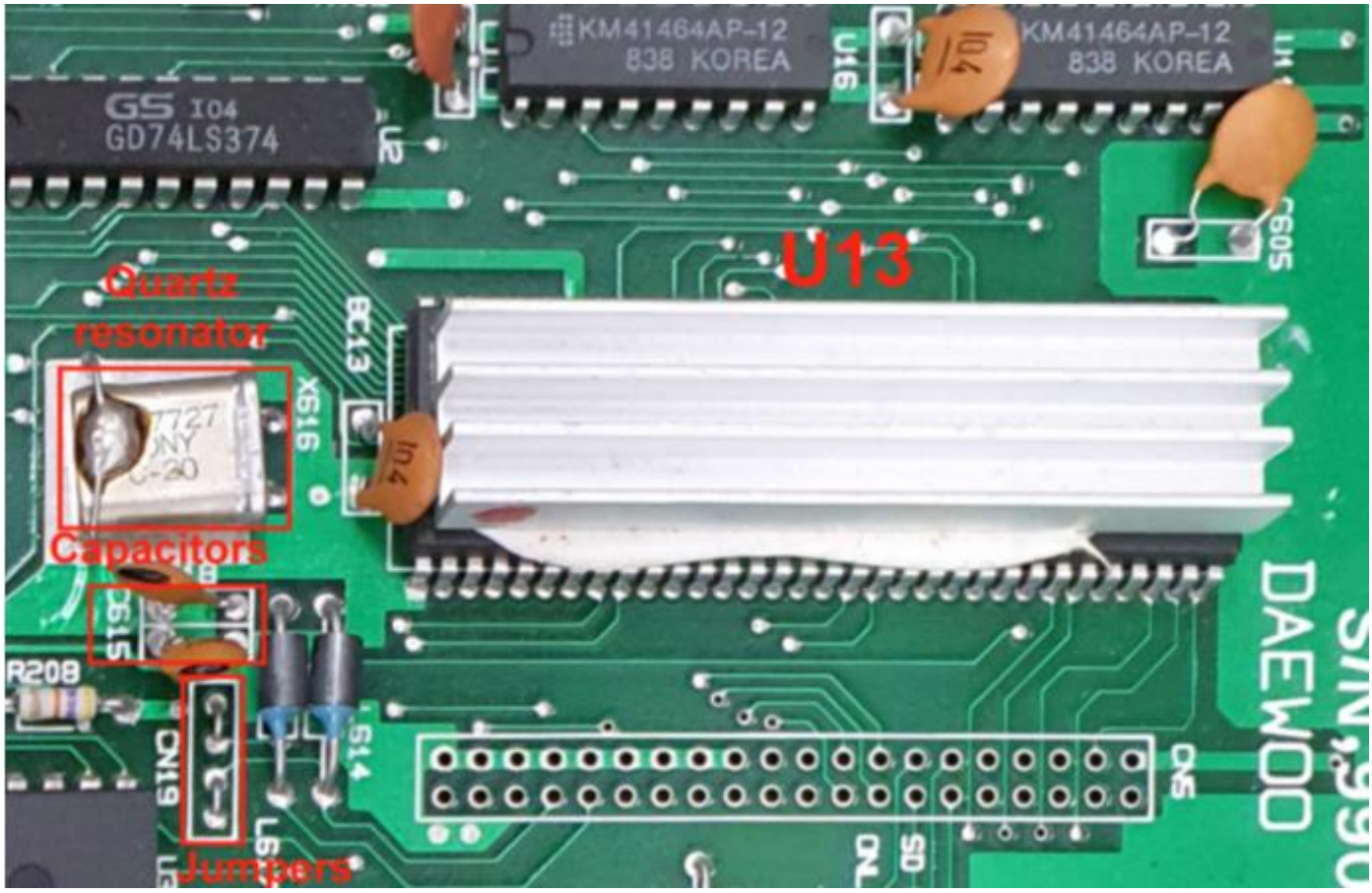
7. Photos of the motherboard

Below are photos of the top side of the motherboard, as well as a photo showing the modifications made to the board for installing the v9958 video processor. Use these photos to locate the necessary components on the board and as a guide for making the modifications.

Photo 1. Location of the main chips: U38 (main ROM), U44 (secondary ROM), U33 (main programmable logic chip), and U46 (additional programmable logic chip):

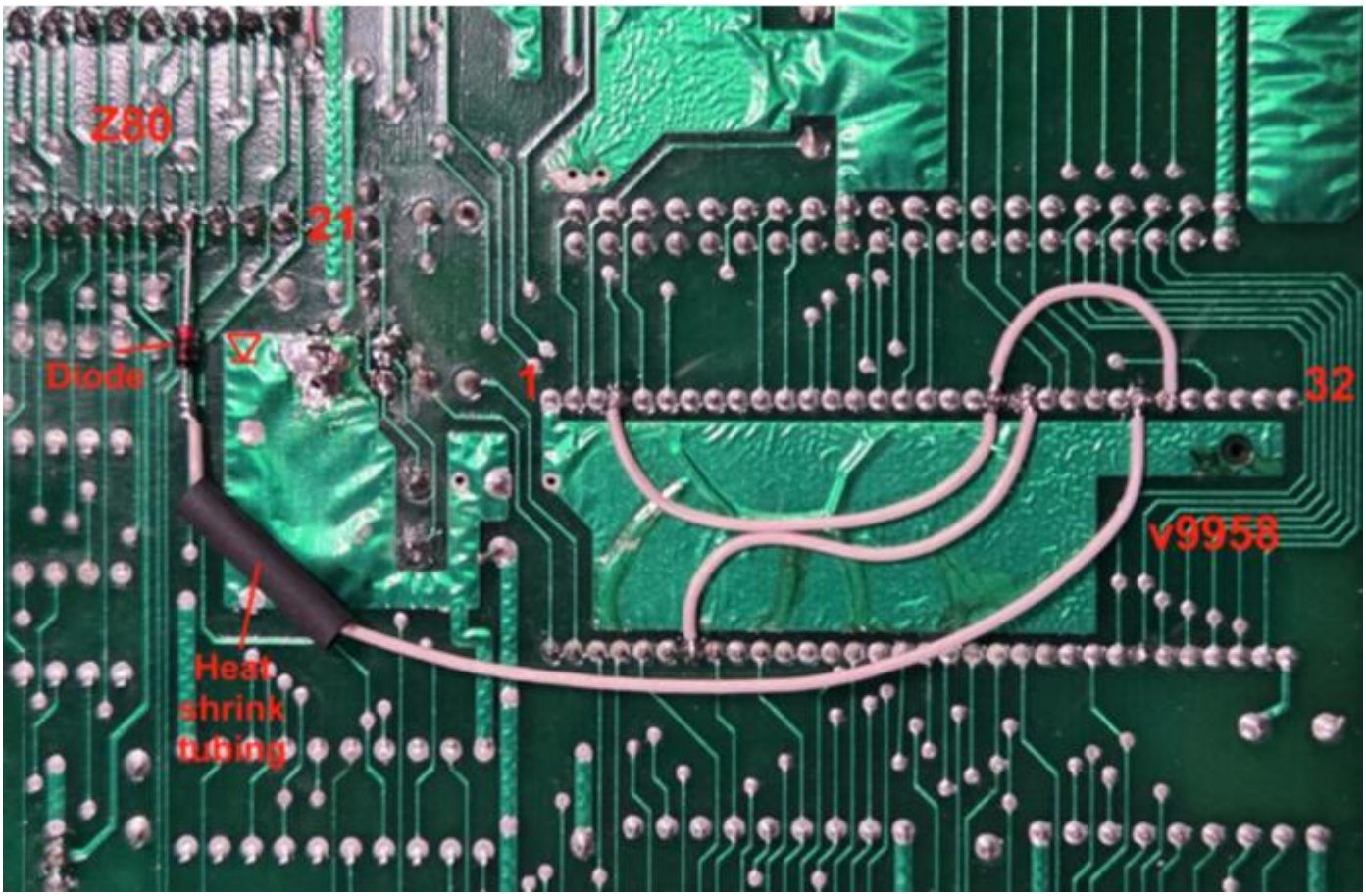
Photo 2. Main RAM (128 KB):

Photo 3. Video processor U13, crystal oscillator, its capacitors, and jumpers:



If, like me, you have a board from a Daewoo 400S and a video board from a 400 model, then you will need to install a 21.47727 MHz crystal oscillator on the board, two 10 pF ceramic capacitors (you may need to select the correct values for your specific crystal oscillator), and two standard jumpers to disable the crystal oscillator if necessary.

Photo 4. Necessary modifications when replacing the video processor with a v9958:



8. Additional information

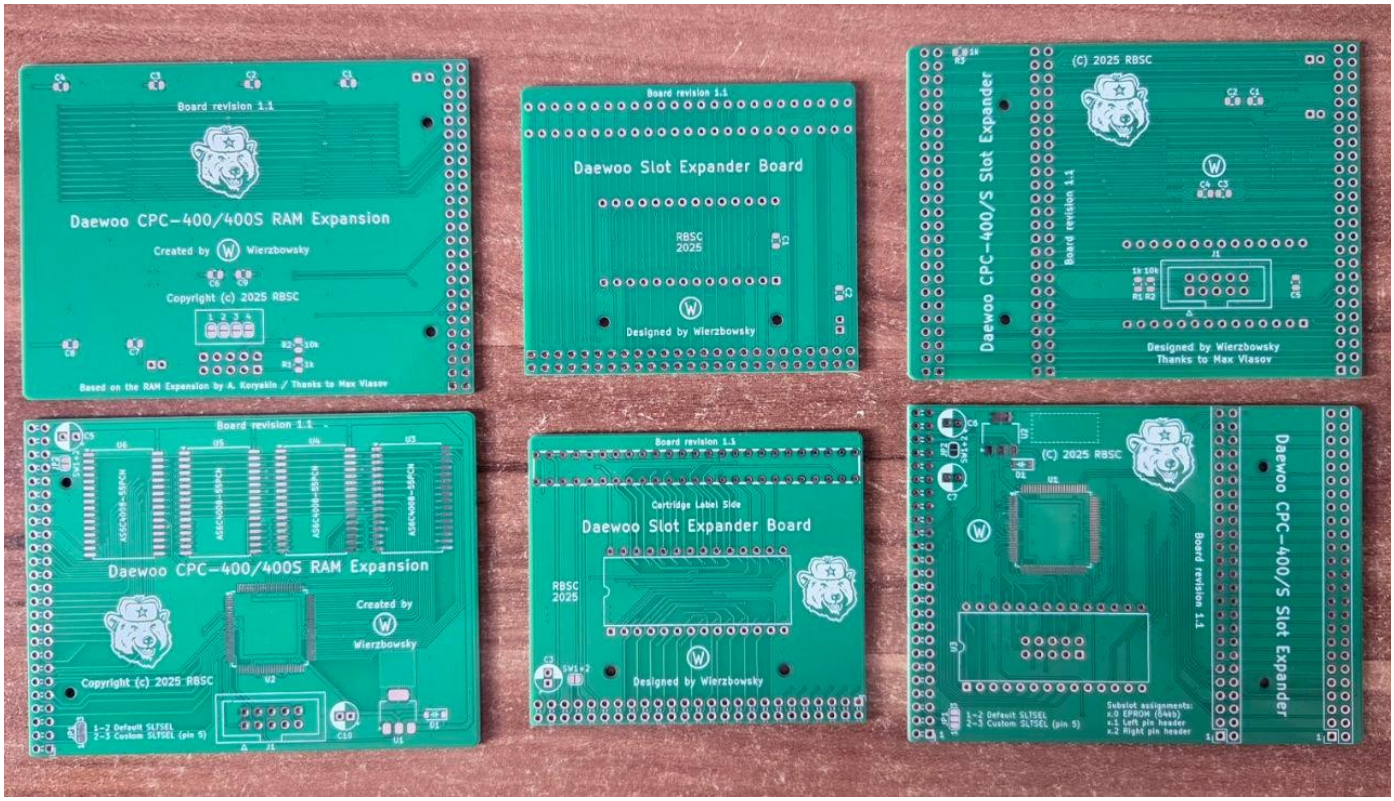
If you don't have a good desoldering pump, such as the ZD-915, or if you lack experience removing chips from a circuit board, it is recommended that you cut them out rather than desolder the RAM chips. These 4464-type DRAM chips are very inexpensive and are still available for purchase. It's better to ruin 4 cheap chips than to ruin the motherboard with a desoldering pump. It is strongly discouraged to desolder these chips with a heat gun—you could overheat and ruin a fairly rare motherboard.

An EPROM programmer, such as the MiniPro TL866, Xgecu T48, or similar, is necessary for writing the required data to ROM chips and GAL programmable logic chips. You can easily buy such a programmer on the Chinese AliExpress; sometimes it is sold in local electronics stores. Depending on the model, it costs $50 or more. There's no need to buy a complete programmer kit with all the adapters; just the programmer and a USB cable for it will suffice. Alternatively, you can find a used programmer or borrow one from an electronics specialist you know. The T48 programmer is recommended, as it's more "up-to-date." The programmers look like this (T48 on the left):

If you are not using electrically erasable Winbond W27C512 EEPROM chips, you will need to purchase UV-erasable 27C512 chips on AliExpress or at your nearest electronics store. To erase them, a special device with an ultraviolet lamp is used, which can easily be purchased for $15. The recommended exposure time for the chips under the UV lamp is 15 minutes. It is advisable to place two layers of regular aluminum foil at the bottom of the chip eraser's drawer. The device for erasing UV chips looks like this:

If you have no prior experience programming EPROM chips or GAL programmable logic devices, we recommend watching a few instructional videos on YouTube to understand how to correctly set the chip type in the programming software, which files to load, and how to verify the correctness of the write operation. This manual does not describe the chip programming methods.

For some upgrade options, you'll need to assemble the boards yourself or purchase pre-assembled ones. You can order the boards from Chinese manufacturers that ship to your country, such as PCBWay or JLCPCB. Five boards (minimum order) with shipping usually cost around $10–15. At a minimum, you need to order the RAM board version 1.1, and it's best to order both the RAM board and the slot expander board. But for a full upgrade, it's recommended to order all 4 boards. The RAM board version 1.0 should only be ordered together with the slot expander board, as this board will conflict with the Daewoo on-board mapper without special firmware in the slot expander. The "bare" upgrade boards look like this (note: the FMPAC board is not shown in the photo):



The repository contains folders with files for manufacturing the boards and even 3D-printed enclosures for them. The Board folder usually contains a file named "board_render.png"—this is a rendering of the board with all components, created in KiCAD.

ATTENTION!

- On boards with solderable jumpers JP1 and JP3, the jumper is soldered between pins 1 and 2. Do not solder jumper JP2
- On the version 1.1 RAM board, it is recommended to install all 4 SRAM chips. In this case, the jumpers on the back of the board do not need to be soldered
- To reduce the available RAM capacity, you can solder the desired jumpers on the back of the RAM board, starting with the fourth one. This will reduce the RAM capacity by 512 KB and disable one of the chips, if it is installed. If jumper 3 is soldered, the RAM capacity will be 1 MB. And so on
- On the FMPAC sound card, you must solder jumpers JP4, JP5, and JP6. If you do not plan to use the stereo audio jack, you do not need to solder jumpers JP4 and JP5
- If an EPROM chip is installed on the board with the MSX slot, you can no longer insert a cartridge into that slot, otherwise there may be a conflict between the two different ROMs

Electronic components for assembling the boards can be ordered on AliExpress or at your local store; a list of required parts is available in the BOM (Bill of Materials) folder for each board. Most components for assembling the boards are available from many sellers, but ordering the **YM2413** FM chip may present some difficulties. Recently, quite a few of these chips have arrived with issues—they either don't work or play music incorrectly. Therefore, it is recommended to order several chips so that at least one turns out to be functional.

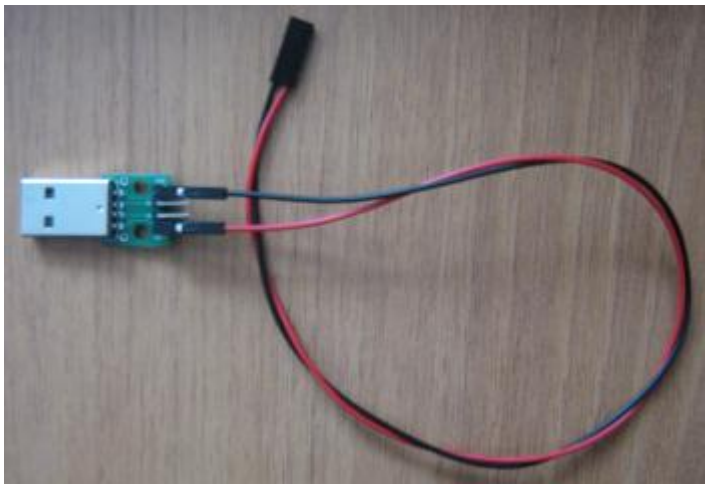
9. Programming boards with CPLDs

The RAM and slot expander boards contain a CPLD chip that needs to be programmed using a USB Blaster programmer, which can also be purchased on AliExpress for just a few dollars. The "Quartus II 15.0 Programmer"

software package is used for flashing, which can be downloaded from here: https:// mirrors. pdp-11. ru/ _ msx/ _ carnivore2// . The USB Blaster programmer looks like this:

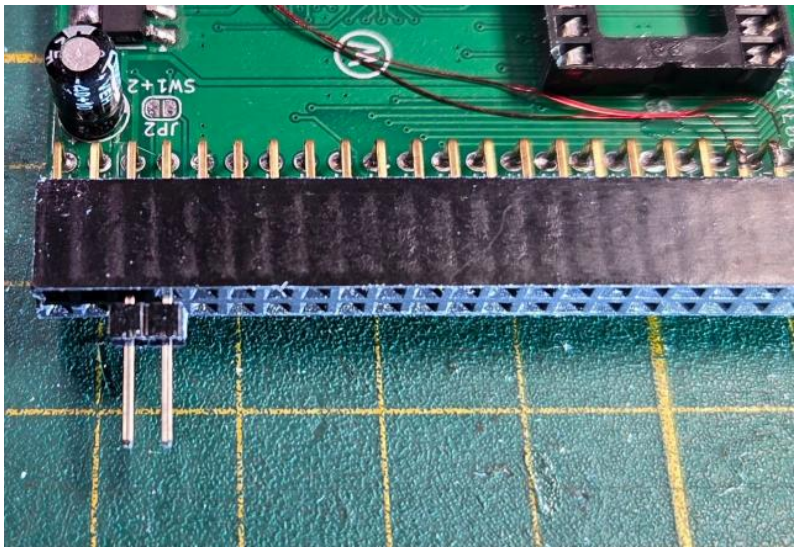
To program CPLDs on RAM boards and slot expansion boards, it is not necessary to solder the JTAG connector. It is sufficient to insert a 2x5-pin header into the "female" connector of the USB Blaster programmer, plug it into the JTAG connector, and bend it slightly for better contact with the board. Examples:

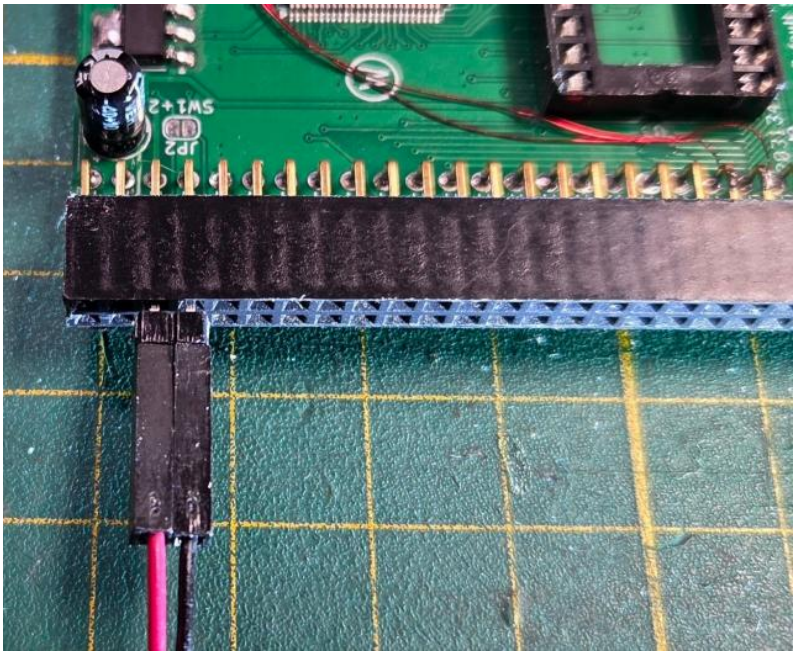
To successfully program the CPLD, you must supply 5 volts of power to the board. This can be done using a standard old USB cable to which two wires with female Dupont-type connectors have been soldered in place of the miniUSB or microUSB connector. Or it can be done using a simple USB adapter assembled from inexpensive Chinese parts, such as this one:



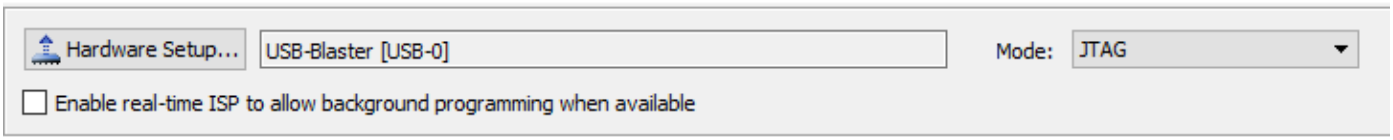
To supply power to the board, insert two long pins into the female connector (this is the connector used to connect the board to the 50-pin rear MSX connector) and connect the adapter or cable described above to it, observing the polarity.

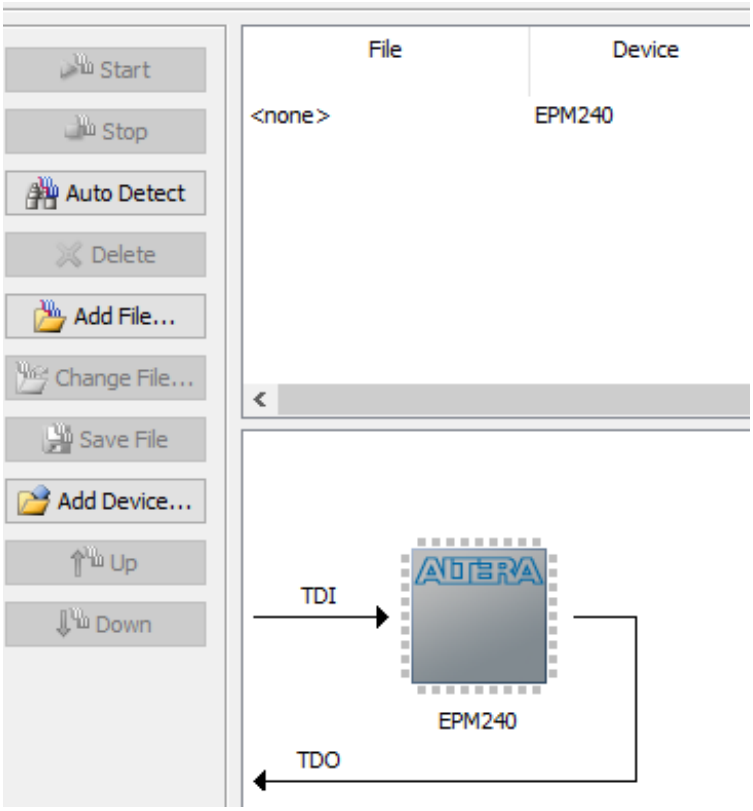
IT IS IMPORTANT TO CONNECT THE POWER SUPPLY CORRECTLY! If you insert the pins incorrectly or reverse the polarity, you may short-circuit the USB power bus, which could damage the power supply. Many modern power supplies have short-circuit protection, but it is still worth checking everything several times before applying power to the board. The power supply pins connect **to the top row of the connector, to the third and fourth pins from the left** when viewed from the connector side (see the below photos). The third pin from the left is the 5-volt power supply, and the fourth pin from the left is "ground." The correct power connection looks like this:



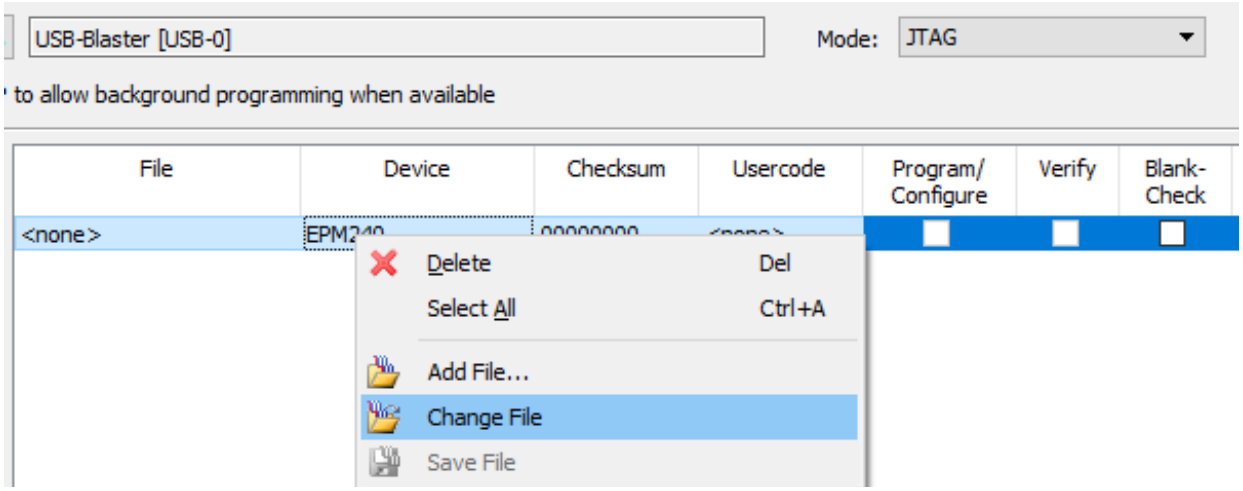


As soon as power is supplied to the board, select JTAG mode in the Quartus software package, connect the USB Blaster programmer connector to the board, and click the Auto-Detect option. If everything is assembled and connected correctly, Quartus will detect the EPM240 chip, as shown in this screenshot:

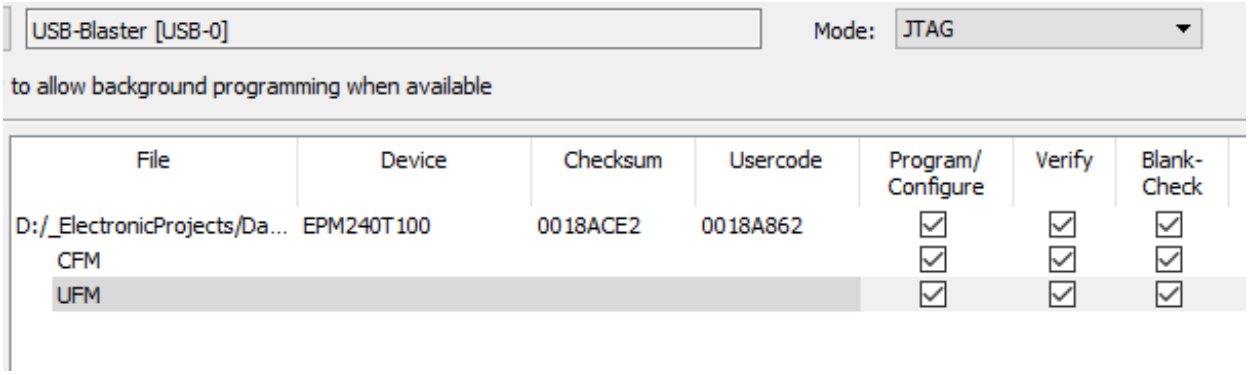




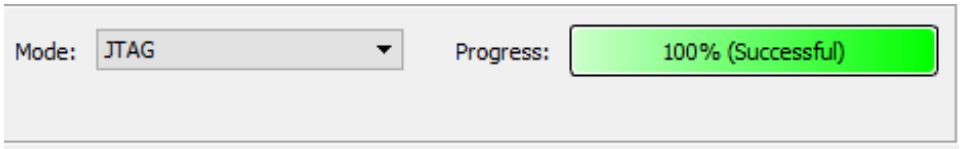
Next, place the cursor on the firmware control (the one with "EPM240" string), right-click, and select the "Change File" option. A file selection window will appear. Locate the firmware you need; it is usually found in the Firmware folder for each CPLD board. Select the .POF file with the desired firmware and open it. Example:



After selecting the firmware file, set the "Program/Configure," "Verify," and "Blank check" options, connect the programmer to the board as recommended above, and click the "Start" button. Example with selected CPLD programming options:



CPLD programming usually takes less than a minute. A sign of successful programming is when the progress bar reaches 100% and no errors appear in the lower window of the interface. Example of successful CPLD programming:




```
Type   ID    Message
   209018 Device 1 silicon ID is ALTERA04-0
   209044 Erasing MAXII/MAXV configuration device(s)
   209019 Blank-checking device(s)
   209023 Programming device(s)
   209021 Performing verification on device(s)
   209011 Successfully performed operation(s)
   209061 Ended Programmer operation at Sat Apr 04 00:34:41 2026
```

Once the CPLD programming is complete, first disconnect the programmer, then disconnect the power cable from the board. The board is now fully ready for use. Install it into the 3D-printed enclosure, secure the board inside the enclosure with short 2mm screws, and connect the two halves of the enclosure with 2x12mm screws. Tighten the screws carefully so as not to damage the plastic.

And one last thing—what's the best way to remove ICs from the sockets? You can't always get a screwdriver in there, and special extractors are either bulky or awkward to use. Lately, we've been using this tool—it's proven itself to be excellent for repairs:

10. Project Repository

This project's repository is located in the main RBSC group repository on GitHub. Project documentation may be updated periodically, so be sure to check that you have downloaded the latest available version.

Permanent link to this repository: https://github.com/RBSC/Daewoo_CPC400_Upgrade

11. Troubleshooting

If the boards you assembled do not work, it is recommended that you thoroughly check everything, reflash the CPLD, and inspect the boards under high magnification for solder bridges, cold solder joints, missing or damaged components, and other anomalies. If you soldered using flux, it must be cleaned off with alcohol.

If your upgrade still fails, you can always revert the Daewoo configuration to its original state by replacing the original U33, U38, U44, and U46 chips, as well as reinstalling the 4 RAM chips (or installing 4 new chips if the old ones were damaged).

We wish you every success with the upgrade! The RBSC group is ready to provide you with all possible assistance in debugging your Daewoo modification, answer any questions about the upgrade, and offer advice on troubleshooting. Our group's contact information can be found in the main project description—in the file "**Daewoo_CPC400_Upgrade_Project_English.pdf**" in this repository.